Ali Darwich Ismail

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SUMMARY

An aspiring robotics engineer with hands-on experience in robotics design, programming, embedded systems, and control systems. Two-time Middle East Champion in the BBC Do Your Bit competition and experienced robotics instructor. Seeking more hands-on experience to contribute to robotics projects in automation, biomechatronics, control, and embedded systems.

EDUCATION

Bachelor of Engineering in Mechatronics Engineering *Sep 2023 - Present*
Lebanese American University, Byblos, Lebanon *Expected Graduation: May 2028*
Lebanese Baccalaureate of General Science *Sep 2020 - Jun 2023*
Rafic Hariri High School, Saida, Lebanon
Graduated with Distinction (16/20)

PROFESSIONAL EXPERIENCE

Robotics Intern at Camerabotics, Beirut, Lebanon *Feb 2026 - Aug 2026*

- Designed a UART communication system between a daisy chain of custom STM32 PCBs and a Raspberry Pi.
- Debugged and extended C++ motor control firmware on STM32, implementing position/velocity control, low-level UART protocol with commands and reply messages, and a flash-based memory map.
- Built the Python motor control API and high-level UART protocol, deployed on a Raspberry Pi.

LAU Robotics Summer Camp, Lead Organizer and Robotics Instructor *Jul 2025, Jul 2024*

- Designed, delivered, and monitored robotics camp sessions for 40+ students using LEGO SPIKE Prime.
- Coordinated camp logistics and managed a team of 4 instructors and 4 volunteers.

PROJECTS

2-DoF Biomimetic Robotic Fish; Project Leader, Embedded System Developer *May 2026*

- Led the team and coordinated the project workflow.
- Developed the inverse kinematics and encoder feedback algorithms and applied them to a cascaded PID controller.

FrameLock: Cinematography Robot with Real-time Face Detection and Tracking; Controller and Raspberry Pi Embedded System Developer, Face Detection and Tracking Developer *November 2025*

- Applied a face detection algorithm using OpenCV on a Raspberry Pi and developed the focal distance algorithm to estimate the distance between the robot and the face.
- Designed and tuned dual PID controllers to stabilize robot orientation and maintain smooth cinematic tracking with switchable subjects.

Ctrl+Farm: An Autonomous Farming Robot; Project Leader, myRIO Embedded System Developer *May 2024*

- Developed the full LabVIEW program for the robot and designed its state machine, sequencing drilling, seeding, watering, and filling, with IMU data for stability monitoring and fall detection.
- Wired and integrated all the electronics on an NI myRIO, including stepper, DC, and servo motors.

TECHNICAL SKILLS

- **Programming**: C/C++, Java, MATLAB/Simulink, Python, ROS2, OpenCV, LabVIEW.
- **Robotics/Embedded Systems**: Sensors and Sensor Fusion, Control Systems, Arduino, myRIO, Raspberry Pi, STM32.
- **Software Tools**: AutoCAD, SolidWorks, Microsoft Office, VS Code, Linux, Git and GitHub.

AWARDS AND CERTIFICATIONS

- *Two-time Middle East Champion, Do Your Bit Competition by BBC (2021, 2022).*
- *Best Programmer, Do Your Bit National Competition (2023).*
- *Programming in C++: A Hands-on Introduction specialization by Codio, Coursera.*
- *Arduino For Beginners course by Packt, Coursera.*
- *Introduction to Programming with MATLAB by Vanderbilt University, Coursera.*
- *Robotics specialization by the University of Pennsylvania, Coursera.*

LANGUAGES

- Arabic (Native)
- English (C2)
- French (B2)